

Missouri University of Science and Technology

Intro to Data Science
CS 3402 (Spring 2026)

Lecture 10 Quiz1 Review & Group Project Topic 2

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Quiz1 Material Review

Content will be covered in the first quiz:

Course Schedule

Week 1	01/20: Introduction to the Course 01/22: Introduction to Data Science ✓
Week 2	01/27: Data Science Tools – Python/Libraries ✓ 01/29: Data Science Tools – IDE and Pytorch ✓
Week 3	02/03: Descriptive Statistics -I ✓ 02/05: Descriptive Statistics -II ✓
Week 4	02/10: Derivatives in Deep Learning/Machine Learning ✓ 02/12: Supervised Learning I ✓ (Homework1 Out)
Week 5	02/17: Supervised Learning II ✓ 02/19: Quiz 1 Material Review

Quiz1 Overview

Topics Covered

- Introduction to Data Science
- Python & Libraries
- IDE / PyTorch basics
- Descriptive Statistics
- Derivatives in Deep Learning
- Supervised Learning

Total: 100 points

Time: 60 minutes

If you finish the quiz 1 less than 1hour, you may go outside for a break.

We will introduce the 3rd group project after the quiz.

Quiz1 Structure

Part I — Data Science Foundations (Conceptual Basics) - Easy

Part II — Python & Tools (Practical Understanding) - Easy-Medium

Part III — Descriptive Statistics - Medium

Part IV — Derivatives in Deep Learning - Medium

Part V — Supervised Learning - Medium

Questions can come in YES/NO, Multiple Choice, Short Answer, and Calculation.

Quiz1 Example Questions

Part I — Data Science Foundations (Conceptual Basics) - Easy

Purpose: Understand terminology and high-level ideas.

1. What is the relationship between Data Science/AI/Machine Learning/Deep Learning?
2. What is supervised learning?
3. Difference between NumPy and PyTorch?
4. Supervised learning requires labeled data. (Yes/No)
5. An IDE helps organize, run, and debug code. (Yes/No)

Quiz1 Example Questions

Part II — Python & Tools (Practical Understanding) - Easy-Medium

Purpose: Understand basic syntax and libraries.

Which library is mainly used for numerical arrays?
A. pandas
B. matplotlib
C. NumPy
D. PyTorch

Which one is a common IDE?
A. Visual Studio Code
B. PyCharm
C. Jupyter Notebook
D. All of the above

PyTorch tensors are similar to:
A. Python dictionaries
B. NumPy arrays
C. Strings
D. HTML files

Python Code Interpreation:
import numpy as np
x = np.array([1,2,3,4])
print(np.mean(x))
What is the output?
What does np.mean() compute?

Quiz1 Example Questions

Part III — Descriptive Statistics - Medium

Purpose: Connect math to data interpretation.

Calculation
Given data:
[2,4,4,6,10]
(a) Compute the mean
(b) Compute the variance
(c) What does variance measure?
(d) What is the median?

If variance is large, it means:
A. Data points are close to the mean
B. Data points are widely spread
C. Mean is large
D. Median equals mean

Which of the following statements is TRUE about a Probability Mass Function (PMF)?
A. It applies only to continuous random variables.
B. The sum of all probabilities equals 1.
C. The area under the curve equals 1.
D. It can take negative values.

A startup tracks the number of customer complaints per day. The probabilities are proportional to the number of complaints:

$$P(X = x) = kx, x = 1,2,3,4$$

(a) Find the value of k, (b) Compute P(X>2), (c) Compute the Variance of X

Quiz1 Example Questions

Part IV — Derivatives in Deep Learning - Medium

Purpose: Light mathematical reasoning. Understand the foundations of how weights update in ANNs.

Basic Derivative:

If $f(x) = x^2 + x$

Compute

$$\frac{df}{dx}$$

Gradient Chain Rule
Assume a and b are constants.

If $f(x) = (bx - a)^2$

Compute

$$\frac{df}{dx}$$

Why are derivatives important in training neural networks?

Expected answer:
Used in gradient descent (Help update model weights)
Help minimize loss

Quiz1 Example Questions

Part V — Supervised Learning - Medium

Purpose: Connect statistics + ML intuition.

What is a loss function?
Give one example.

A model gets:
99% training accuracy
60% test accuracy
Do you think this is a good model and Why?
What is happening?

Gradient Chain Rule

Gradient Chain Rule

If

$$L = L(\hat{y}), \quad \hat{y} = f(\mathbf{w}, \mathbf{x})$$

Then for each weight w_j:

$$\boxed{\frac{\partial L}{\partial w_j} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w_j}}$$

This is the **gradient chain rule**.

Model

$$\hat{y} = w_1 x_1 + w_2 x_2$$

Loss

$$L = \frac{1}{2}(\hat{y} - y)^2$$

Step-by-step chain rule

Step 1: Loss → prediction

$$\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$$

Step 2: Prediction → weights

$$\frac{\partial \hat{y}}{\partial w_1} = x_1, \quad \frac{\partial \hat{y}}{\partial w_2} = x_2$$

Step 3: Chain them

$$\frac{\partial L}{\partial w_1} = (\hat{y} - y)x_1$$

$$\frac{\partial L}{\partial w_2} = (\hat{y} - y)x_2$$

That's exactly what we used in gradient descent.

Gradient Chain Rule

Model

$$\hat{y} = w_1 x_1 + w_2 x_2$$

Loss

$$L = \frac{1}{2}(\hat{y} - y)^2$$

Step-by-step chain rule

Step 1: Loss → prediction

$$\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$$

Step 2: Prediction → weights

$$\frac{\partial \hat{y}}{\partial w_1} = x_1, \quad \frac{\partial \hat{y}}{\partial w_2} = x_2$$

Step 3: Chain them

$$\frac{\partial L}{\partial w_1} = (\hat{y} - y)x_1$$

$$\frac{\partial L}{\partial w_2} = (\hat{y} - y)x_2$$

That's exactly what we used in gradient descent.

Step 4: Substitute related terms into results.

$$\frac{\partial L}{\partial w_1} = (w_1 x_1 + w_2 x_2 - y)x_1$$

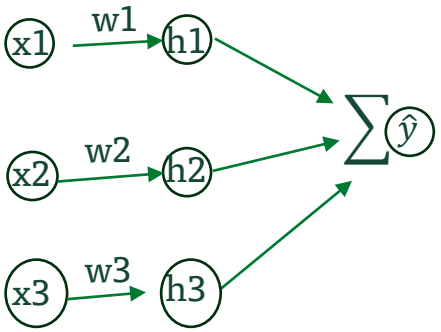
$$\frac{\partial L}{\partial w_2} = (w_1 x_1 + w_2 x_2 - y)x_2$$

Quiz1 Example Questions

Part V — Supervised Learning - Medium

Purpose: Connect statistics + ML intuition.

Update ANN model weights.



Initial weights: $w1 = 1, w2 = 1, w3 = 1$

Learning rate: 0.1

Loss function (squared error): $\mathcal{L} = \frac{1}{2}(\hat{y} - y)^2$

Two Training samples: [(2, 1, 1),4] and [(0, 1, 3), 2]

(x1, x2, x3)

y

(a) Formulate the model

Update the weights using the two training samples. Learning rate is unchanged during training.

(b) Please use the first training sample at the first iteration. Calculate the updated weights after first iteration.

(c) Use the second training sample at the second iteration. Calculate the updated weights after second iteration.

Toy Example of Gradient Descent in ML Models

Model: $\hat{y} = f(x_1, x_2, x_3) = w_1x_1 + w_2x_2 + w_3x_3$

Two Training samples: : $[(2, 1, 1), 4]$ and $[(0, 1, 3), 2]$

Loss function (squared error): $\mathcal{L} = \frac{1}{2}(\hat{y} - y)^2$

Learning rate η : *0.1*

Initial weights: $w1 = 1, w2 = 1, w3 = 1$

Step 1: Compute the prediction

Step 2: Compute the Loss

Step 3: Compute the gradients

Step 4: Gradient descent update rule $w_i^{new} \leftarrow w_i^{old} - \eta \frac{\partial \mathcal{L}}{\partial w_i}$

For the 1st iteration.

Step1: $\hat{y} = w_1x_1 + w_2x_2 + w_3x_3 = 1 * 2 + 1 * 1 + 1 * 1 = 4$

Step2: $\mathcal{L} = \frac{1}{2}(\hat{y} - y)^2 = \frac{1}{2}(4 - 4)^2 = 0$

Step 3: $\frac{\partial \mathcal{L}}{\partial w_1} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w_1} = (\hat{y} - y)\, x_1 = 0$ $\frac{\partial \mathcal{L}}{\partial w_2} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w_2} = (\hat{y} - y)\, x_2 = 0$ $\frac{\partial \mathcal{L}}{\partial w_3} = = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w_3} = (\hat{y} - y)\, x_3 = 0$

Step 4: $w_1^{new} = w_1^{old} - \eta \frac{\partial \mathcal{L}}{\partial w_1} = 1 - 0.1 * 0 = 1$ $w_2^{new} = w_2^{old} - \eta \frac{\partial \mathcal{L}}{\partial w_2} = 1 - 0.1 * 0 = 1$

$w_3^{new} = w_3^{old} - \eta \frac{\partial \mathcal{L}}{\partial w_3} = 1 - 0.1 * 0 = 1$

Toy Example of Gradient Descent in ML Models

Model: $\hat{y} = f(x_1, x_2, x_3) = w_1x_1 + w_2x_2 + w_3x_3$

Two Training samples: : $[(2, 1, 1), 4]$ and $[(0, 1, 3), 2]$

Loss function (squared error): $\mathcal{L} = \frac{1}{2}(\hat{y} - y)^2$

Learning rate η : *0.1*

Initial weights: $w1 = 1, w2 = 1, w3 = 1$

Step 1: Compute the prediction

Step 2: Compute the Loss

Step 3: Compute the gradients

Step 4: Gradient descent update rule $w_i^{new} \leftarrow w_i^{old} - \eta \frac{\partial \mathcal{L}}{\partial w_i}$

For the 2nd iteration.

Step1: $\hat{y} = w_1x_1 + w_2x_2 + w_3x_3 = 1 * 0 + 1 * 1 + 1 * 3 = 4$

Step2: $\mathcal{L} = \frac{1}{2}(\hat{y} - y)^2 = \frac{1}{2}(4 - 2)^2 = 2$

Step 3: $\frac{\partial \mathcal{L}}{\partial w_1} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w_1} = (\hat{y} - y)\, x_1 = 0$ $\frac{\partial \mathcal{L}}{\partial w_2} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w_2} = (\hat{y} - y)\, x_2 = 2$ $\frac{\partial \mathcal{L}}{\partial w_3} = = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w_3} = (\hat{y} - y)\, x_3 = 6$

Step 4: $w_1^{new} = w_1^{old} - \eta \frac{\partial \mathcal{L}}{\partial w_1} = 1 - 0.1 * 0 = 1$ $w_2^{new} = w_2^{old} - \eta \frac{\partial \mathcal{L}}{\partial w_2} = 1 - 0.1 * 2 = 0.8$

$w_3^{new} = w_3^{old} - \eta \frac{\partial \mathcal{L}}{\partial w_3} = 1 - 0.1 * 6 = 0.4$

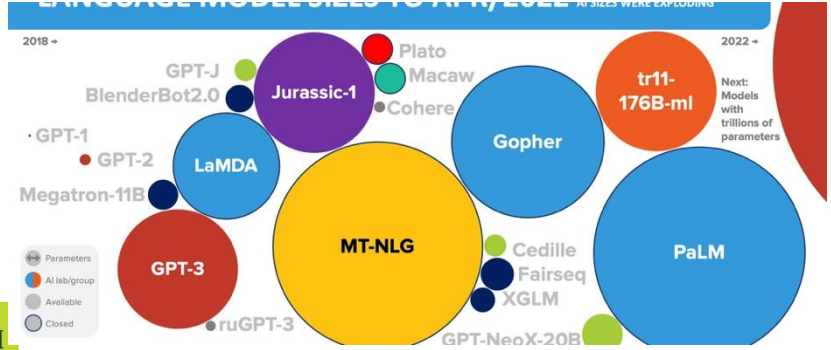
Group Project Topic 2

Title: Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

Background: Large Language Models (LLMs) such as GPT and Gemini are trained on massive text corpora collected from the internet. However, these training datasets are heavily dominated by English and high-resource regions, potentially leading to uneven performance across languages and countries. As a result, LLMs may produce responses that vary in accuracy, completeness, reasoning quality, or depth depending on the language or regional context of the prompt.

Group Project Topic 2

Aim: This project aims to systematically evaluate whether LLM performance is consistent across different languages (and potentially across countries or regions). By designing controlled prompts and comparing model responses using quantitative metrics such as accuracy, length, sentiment, and reasoning quality, students will analyze whether measurable discrepancies exist. This project highlights important issues in data representation, model generalization, and fairness in AI systems.





Group Project Topic 2

In this project, you will explore whether **large language models (LLMs)** perform consistently across different languages and regional contexts. You will design controlled prompts (e.g., factual questions, reasoning tasks, or knowledge-based queries) and evaluate model responses in multiple languages and/or across references to different countries. At least two languages (or regions) should be included to enable systematic comparison. The goal is to quantitatively assess differences in response accuracy, completeness, reasoning quality, sentiment, or other measurable characteristics, and to determine whether observable performance discrepancies exist.

Through this investigation, you will gain hands-on experience in experimental design for AI evaluation, prompt construction, quantitative metric development, and statistical comparison. More broadly, you will develop critical insight into how data distribution, representation imbalance, and training corpora composition may influence model behavior, robustness, and fairness in global AI systems.

Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

1. Data Collection and Preprocessing

You should design a structured prompt dataset to evaluate LLM performance across **at least two** languages (and optionally across different countries or regions). Prompts may include factual knowledge questions, reasoning tasks (e.g., math or logic), short explanatory questions, or domain-specific queries. Public question sources such as multilingual QA datasets, Wikipedia-based fact queries, etc.

Dataset description: Task type (factual QA, reasoning, explanation, sentiment, etc.); Languages included (e.g., English, Spanish, Chinese, etc.); Optional regional/country references (if applicable); Source of prompts (public dataset, translated version, or self-constructed).
Data inspection and quality check: Ensure semantic equivalence across languages (careful translation or back-translation if necessary); Remove ambiguous or culturally dependent questions that may introduce confounding factors; Verify correctness of reference answers (for objective tasks);
Data preprocessing: Ensure consistent instructions across languages; If translation is used, clearly document the translation method (human vs. automated).
Dataset statistics: Number of task categories; Number of prompts per language;

Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

2. Model Selection and Experimental Design

You must evaluate multiple large language models (LLMs) across multiple languages (and optionally across regional or country-based prompts) under controlled experimental settings.

(a) Pretrained Large Language Models (LLMs) - Closed-source models

You should select widely used pretrained foundation model of closed-source. Comparing multiple models is encouraged.

Examples: **ChatGPT, Claude, Gemini,** or similar models

Learning settings. You should evaluate model performance under consistent and clearly defined settings:

Zero-shot learning: no task-specific examples provided

Few-shot learning: a small number of labeled examples provided as prompts

To ensure a fair comparison across languages or regions, you must clearly describe:

Prompt design: Exact prompt templates used. Method of translation (manual, professional, automated).

Number of prompts: Total prompts per language. Number of prompts per task type.

Evaluation method: Whether human evaluation is used (if applicable). Whether responses are evaluated automatically (e.g., exact match, BLEU, accuracy).

Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

2. Model Selection and Experimental Design

You must evaluate multiple large language models (LLMs) across multiple languages (and optionally across regional or country-based prompts) under controlled experimental settings.

(a) Pretrained Large Language Models (LLMs) - Open-source models

You should select widely used pretrained foundation model of open source models. Comparing multiple models is encouraged.

Examples: **Open-source LLMs** (e.g., LLaMA-based models, Falcon-40B, Mistral) or similar models

Learning settings. You should evaluate model performance under consistent and clearly defined settings:

Zero-shot learning: no task-specific examples provided

Few-shot learning: a small number of labeled examples provided as prompts

To ensure a fair comparison across languages or regions, you must clearly describe:

Prompt design: Exact prompt templates used. Method of translation (manual, professional, automated).

Number of prompts: Total prompts per language. Number of prompts per task type.

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Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

2. Model Selection and Experimental Design

You must evaluate multiple large language models (LLMs) across multiple languages (and optionally across regional or country-based prompts) under controlled experimental settings.

(b) Fine-Tuning Small Large Language Models

In addition to pretrained models, you are required to fine-tune at least one smaller open-source LLM (e.g., **GPT-2**, **LLaMA-1B**, or similar) on a subset of the prompt dataset.

- Collect at least one text dataset appropriate for your task (factual QA, reasoning, or explanatory prompts).
- Adapt the model to the selected task using the training subset.
- Reserve a separate set of prompts not used in fine-tuning to enable fair performance comparison.
- Clearly report all hyperparameters and settings, including learning rate, batch size, number of epochs, optimizer, and any data preprocessing applied.

Goal: Compare the performance of the fine-tuned small LLM with zero- or few-shot pretrained LLMs to analyze how task-specific adaptation affects accuracy, reasoning, and cross-linguistic consistency.

Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

3. Evaluation and Analysis

For all models (pretrained and fine-tuned):

Prompt design: clearly specify templates, translations, and examples used.
Number of prompts: total prompts per language, per task.
Metrics: accuracy, BLEU/ROUGE, response length, reasoning quality, sentiment, etc.
Statistical comparison: paired tests to quantify cross-linguistic performance differences.

Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

4. Documentation, Reporting, and Presentation

4.1 Project Report

You must submit a written report that includes:

- Introduction and motivation
- Dataset description and preprocessing
- Model descriptions and experimental setup
- Results and comparisons
- Discussion and key findings
- Limitations and future directions
- All results should be clearly organized using tables and figures.

Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

4. Documentation, Reporting, and Presentation

4.2 Presentation Slides

You should prepare a slide presentation covering:

- Problem definition and dataset
- Model comparison strategy
- Key experimental results
- Main insights and conclusions
- Presentations should emphasize **what was learned**, not just which model performed best.

Peer Evaluation of Presentations

- In addition to instructor evaluation, each team's presentation will be **rated by other teams** based on its **overall quality, clarity, and organization**.
- Please ensure your slides are well-structured and easy to follow, Clear visuals and a logical flow are strongly encouraged.